

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 30

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Booms are floating tubes with a skirt that restrict the movement of floating oil, keeping it near the source / away from sensitive sites (mangroves, reef, marine park) • They work best in sheltered / calm water such as the lagoon • Limitation: they do not work in open water where currents or waves are strong; oil can flow under the skirt in a strong current	3
01.2	• Skimmers: rotating (metal) discs / belts pick up floating oil which is scraped off and stored • Absorbent materials (high surface area, eg textile mops, straw, human hair) soak up oil and are then removed • Collected oil / absorbents are incinerated or sent to landfill (or oil recycled)	3
01.3	• Dispersants break the oil into an emulsion of droplets, making it more mobile so it spreads through the water column (to the reef / seabed organisms) rather than staying on the surface • They do not reduce the amount of oil; the dispersant / oil mixture may be toxic to corals and plankton in shallow, enclosed water	2
01.4	• Bacteria that break down hydrocarbons digest the residual oil (most rapid where temperature, oxygen and nutrients are high) • Accelerated by adding nutrients (nitrates / phosphates) and aerating (ploughing) / irrigating the sediment	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 261-262; AQA spec 3.4.3.2.7.	
02.1	• $13 / 480 = 0.0271$ years • 0.0271×365 • = 10 days (accept 9.9-10)	3
02.2	• $1\,340\,000 / 425 = 3153$ years (accept 3100-3200)	1

Question	Marking guidance	Marks
02.3	• It has a very short residence time / very rapid transfer rate ($480 \times 10^{15} \text{ dm}^3$ per year) • So it moves large volumes of water each year, delivering precipitation to land; small reservoirs with fast transfers respond quickly to change	2
02.4	• The ocean is so large that an added or removed volume is a tiny proportion of the total • The same volume is a large proportion of a small reservoir so its level / residence time changes noticeably	2
02.5	• Water is constantly moving between reservoirs (evaporation, precipitation, runoff, infiltration, transpiration) • But inflows and outflows for each reservoir balance so the volume in each reservoir stays constant over long periods	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 115-116; AQA spec 3.2.2.1.	
03.1	<i>Any 3 from:</i> • Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) • Very complex, expensive technology needing an advanced industrial infrastructure • Concerns about links between civil nuclear materials and weapons / terrorism • Uncertainty over permanent disposal of radioactive waste • Uncertain decommissioning costs	3
03.2	• Only 0.7% of natural uranium is fissile U-235; the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) • Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) • So much more energy is harnessed from the mined uranium; fast reactors need no moderator but are complex and costly	3
03.3	• Advantages: thorium is three times more abundant; much less radioactive waste with shorter half-lives; harder to make weapons material; no fuel enrichment needed • Disadvantages: thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate); U-233 is an alpha emitter so very hazardous to handle; less developed technology with high development costs	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	
04.1	• The energy used to extract raw materials, manufacture the components and assemble the vehicle (for a typical car about 1.5 years of its fuel use) • Reduced by using recycled materials (eg recycled aluminium needs 1/20 of the energy of new) • Lighter designs using less material; longer vehicle life spreading the embodied energy	3
04.2	<i>Any 4 from:</i> • Use of recyclable materials wherever possible • Easy identification of components and their composition (code stamps) • Easy dismantling and separation of components / materials • Reusable components for new vehicles • Compostable materials for parts that cannot be recycled	4

Question	Marking guidance	Marks
04.3	- At low speed a large proportion of the energy just runs the engine rather than moving the car - At high speed energy is wasted overcoming air resistance (which rises with the square of speed); optimum for many cars about 56 mph	2
04.4	Automatic stop / start systems cutting the engine when stationary; smooth driving in higher gears avoiding sudden braking / acceleration	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 227-228; AQA spec 3.3.4.2.1.	
05.1	- Ploughing: increases the oxygen supply to soil decomposers so aerobic decomposition of dead organic matter is faster; carbon in the soil DOM store falls and atmospheric CO₂ rises - Draining: waterlogged anaerobic conditions preserved dead organic matter (peat) and produced methane slowly; drainage makes the soil aerobic so stored carbon is rapidly decomposed to CO₂ (though methane release stops) - Marine pollution: kills phytoplankton so less dissolved CO₂ is absorbed by photosynthesis and less carbon sinks to sediments in dead plankton; less CO₂ then dissolves from the atmosphere to replace it, so atmospheric CO₂ rises	6
05.2	Any 3 from: - Rice paddy fields - Landfill sites - Anaerobic sediments in reservoirs - Livestock intestines (cattle) - Manure / slurry stores	3
05.3	More CO₂ dissolves in the sea, forming carbonic acid which dissociates into hydrogen carbonate and hydrogen ions; more H⁺ lowers the pH	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 150-151; AQA spec 3.2.4.2.	
06.1	The readings are precise but may not be accurate	1
06.2	- A result that differs from other results with which it was expected to be similar - Repeat the study / measurement: if it recurs it is a 'real' result caused by another variable; if not it was an error and can be excluded	2
06.3	- To test the reliability of the method / equipment - To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
07.1	- Overgrazing: vegetation is eaten faster than it grows so soil is exposed; hooves disturb the soil, damage roots and compact it, reducing infiltration and increasing runoff - Reduced biota (ploughing, agrochemicals, less organic matter): less humus so less adhesion between particles; fewer worms so less aeration and lower infiltration capacity; fewer nutrients released so less vegetation cover	4

Question	Marking guidance	Marks
07.2	• Tree roots that held the soil together decompose so the soil is bound less strongly • Heavy rain adds mass and lubricates particles; without interception more water enters the soil, making slumping more likely	2
07.3	<i>Any 4 from:</i> • Sedimentation in rivers reduces flow capacity so flooding downstream is more likely • Sedimentation in reservoirs reduces their storage volume • Coastal sedimentation covering reefs / seagrass; turbidity reducing light for aquatic plants • Increased atmospheric particulates from wind erosion worsening asthma; dust covering crops / urban areas • Desertification as vegetation cannot re-establish; reduced rainfall • Nutrients (phosphates) carried on soil particles causing eutrophication	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	
08.1	• Doubling the distance from 1 m to 2 m reduces the dose rate from 400 to 100 (one quarter) • Distance x 10 (1 to 10 m) reduces it by 100 times (400 to 4); dose is proportional to $1/\text{distance}^2$	2
08.2	• $400 / 20^2 = 1 \mu\text{Sv h}^{-1}$	1
08.3	• Sealed container prevents contact / contamination so exposure stops when the worker moves away; container may absorb alpha completely; remote handling • Barriers (lead, concrete, thick steel flask walls) absorb radiation before it reaches the worker; thickness depends on radiation type • Dose received is proportional to time near the source so work is organised to minimise time close to it (job rotation) • Dosimeters / film badges record exposure; air monitors detect alpha-emitting particles; contamination monitors on leaving check that no material is carried out	4
08.4	• Stable iodine saturates the thyroid • So radioactive iodine-131 released is not taken up / concentrated in the thyroid gland	2
08.5	• Exposure should be As Low As Reasonably Achievable	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 302-303; AQA spec 3.4.3.2.14.	
09.1	• Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated • Grit traps: the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them	2
09.2	• The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge • This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced	2

Question	Marking guidance	Marks
09.3	• Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter • Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process • The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge'; returning them maintains a large bacterial population so digestion is rapid	3
09.4	• Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria • Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) • Phosphate stripping with iron(III) sulfate; micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9

Question	Marking guidance	Marks
10.2	• The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25

Question	Marking guidance	Marks
11.2	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25